

# CCNA 200-301 - Video 03: Network Fundamentals (Part 2)

## Key Topics Covered

- NIC (Network Interface Card / Network Adapter).
- Ethernet speeds: Ethernet (10 Mbps), Fast Ethernet (100 Mbps), Gigabit Ethernet (1000 Mbps).
- MAC Address (Media Access Control Address).
- Hub vs Switch.
- Signal Types: Simplex, Half Duplex, Full Duplex.
- Casting Types: Unicast, Broadcast, Multicast.

## Important Notes

- Every network device needs a NIC to communicate over the network.
- Each NIC has a unique MAC Address used at Layer 2.
- The first part of a MAC identifies the manufacturer (OUI); the remaining part identifies the device.
- Hub works at Layer 1 and sends traffic to all ports.
- Switch works at Layer 2 and forwards frames based on MAC Address.
- Half Duplex: transmit OR receive at one time.
- Full Duplex: transmit AND receive simultaneously.
- Unicast = one-to-one, Broadcast = one-to-all, Multicast = one-to-many.

## Hub vs Switch

| Feature      | Hub         | Switch                          |
|--------------|-------------|---------------------------------|
| OSI Layer    | Layer 1     | Layer 2                         |
| Transmission | Half Duplex | Full Duplex                     |
| Traffic      | Broadcast   | Unicast / Multicast / Broadcast |

## Exam Tips

- Remember: Switch = Layer 2, Hub = Layer 1.
- MAC Address belongs to the NIC.
- Know the difference between Unicast, Broadcast and Multicast.
- Know the difference between Half Duplex and Full Duplex.

## Cisco Commands

No Cisco IOS commands were introduced in this lesson.